Integral tak Tentu

DATE :

Latihan 1

A

1. $\int (x^4 - 3x^3 + 5x^2 - 11x + 4)\,dx$
2. $\int (4\sqrt[3]{x} - \frac{3}{\sqrt{x}})\,dx$
3. $\int (\frac{5}{x^2} + \frac{4}{x^3} - \frac{6}{x^4})\,dx$
4. $\int (\frac{4x^3 - 3x - 1}{x^4})\,dx$
5. $\int (\frac{t + 2t^2}{\sqrt{t}})\,dt$

Jawab

1. $\int x^4 dx - \int 3x^3 dx + \int 5x^2 dx - \int 11x\,dx + \int 4\,dx$

$\frac{1}{5}x^5 - \frac{3}{4}x^4 + \frac{5}{3}x^3 - \frac{11}{2}x^2 + 4x + C$ //

2. $\int (4x^{1/3} - 3x^{-1/2})\,dx$

$\int 4x^{1/3}\,dx - \int 3x^{-1/2}\,dx$

$\frac{4}{4/3}x^{4/3} - \frac{3}{1/2}x^{1/2} + C = 3x^{4/3} - 6x^{1/2} + C$ // $= 3x\sqrt[3]{x} - 6\sqrt{x} + C$ //

3. $\int (5x^{-2} + 4x^{-3} - 6x^{-4})\,dx$

$\int 5x^{-2}\,dx + \int 4x^{-3}\,dx - \int 6x^{-4}\,dx$

$\frac{5}{-1}x^{-1} + \frac{4}{-2}x^{-2} - \frac{6}{-3}x^{-3} + C = -5x^{-1} - 2x^{-2} + 2x^{-3} + C$ //

$= \frac{-5}{x} - \frac{2}{x^2} + \frac{2}{x^3} + C$ //

4. $\int (4x^{-2} - 3x^{-3} - x^{-4})\,dx$

$\int 4x^{-2}\,dx - \int 3x^{-3}\,dx - \int x^{-4}\,dx$

$\frac{4}{-1}x^{-1} - \frac{3}{-2}x^{-2} - \frac{1}{-3}x^{-3} + C$ // $= \frac{-4}{x} + \frac{3}{2x^2} + \frac{1}{3x^3} + C$ //

5. $\int (t^{1/2} + 2t^{3/2})\,dt$

$\int t^{1/2}\,dt + \int 2t^{3/2}\,dt$

$\frac{1}{3/2}t^{3/2} + \frac{2}{5/2}t^{5/2} + C$ // $= \frac{2}{3}t^{3/2} + \frac{4}{5}t^{5/2} + C$ // $= \frac{2}{3}t\sqrt{t} + \frac{4}{5}t^2\sqrt{t} + C$ //

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6 | 6 | $\int \frac{x^3}{(1+x^4)^2}\,dx$

7 | $\int \frac{x}{(x^2-1)^3}\,dx$

8. | $\int x^2\sqrt{x^3+1}\,dx$

9 | $\int \cos^2 x \sin x\,dx$

10 | $\int \frac{\sin x}{\cos^2 x}\,dx$

Jawab

6 | misal $1+x^4 = u \rightarrow \frac{du}{dx} = 4x^3 \rightarrow dx = \frac{1}{4x^3}\,du$

$\int \frac{x^3}{u^2} \cdot \frac{1}{4x^3}\,du$

$\int \frac{1}{4u^2}\,du$

$\frac{1}{4}\int u^{-2}\,du = \frac{1}{4}\left(\frac{1}{-1}u^{-1}\right)+c$

$= -\frac{1}{4u} + c = -\frac{1}{4(1+x^4)} + c //$

7 | misal $x^2-1 = u \rightarrow \frac{du}{dx} = 2x \rightarrow dx = \frac{1}{2x}\,du$

$\int \frac{x}{u^3} \cdot \frac{1}{2x}\,du$

$\frac{1}{2}\int u^{-3}\,du = \frac{1}{2}\left(\frac{1}{-2}u^{-2}\right)+c$

$= -\frac{1}{4u^2} + c = -\frac{1}{4(x^2-1)^2} + c //$

8. | misal $x^3+1 = u \rightarrow \frac{du}{dx} = 3x^2 \rightarrow dx = \frac{1}{3x^2}\,du$

$\int x^2 (u)^{1/2} \cdot \frac{1}{3x^2}\,du = \frac{1}{3}\int u^{1/2}\,du$

$= \frac{1}{3}\left(\frac{1}{3/2}u^{3/2}\right)+c$

$= \frac{2}{9}(x^3+1)^{3/2} + c // = \frac{2}{9}(x^3+1)\sqrt{x^3+1} + c$

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9. $\int \cos^2 x \sin x \, dx$

misal $\cos x = u \rightarrow \frac{du}{dx} = -\sin x \rightarrow dx = \frac{1}{-\sin x} du$

$\int u^2 \cancel{\sin x} \cdot \frac{1}{-\cancel{\sin x}} du$

$-\int u^2 du = -\left(\frac{1}{3} u^3\right) + c = -\frac{1}{3} \cos^3 x + c$ //

10. $\int \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} dx$

$\int \tan x \sec x \, dx = \sec x + c$ //

atau

misal $\cos x = u \rightarrow \frac{du}{dx} = -\sin x \rightarrow dx = \frac{1}{-\sin x} du$

$\int \frac{\cancel{\sin x}}{u^2} \cdot \frac{1}{-\cancel{\sin x}} du$

$-\int \frac{1}{u^2} du = -\int u^{-2} du = -\left(\frac{1}{-1} u^{-1}\right) + c$

$= \frac{1}{u} + c = \frac{1}{\cos x} + c = \sec x + c$ //

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Latihan 5

1. a. $f'(x) = 6x^2 - 4x + 3$ dan $f(-1) = 3$

b. $f'(x) = 6\sqrt{x} - \frac{10}{\sqrt{x}}$ dan $f(4) = 2$

c. $f'(x) = 4\sin 2x + 6\cos 3x$ dan $f(0) = -4$

d. $f''(x) = 12x^2 - 6x + 2$, $f'(0) = 1$ dan $f(2) = 11$

a $\int f'(x) = \int (6x^2 - 4x + 3)\,dx$

$= \int 6x^2 dx - \int 4x\,dx + \int 3\,dx$

$= \frac{6}{3}x^3 - \frac{4}{2}x^2 + 3x + c$

$\int f'(x) = 2x^3 - 2x^2 + 3x + c$

$f(-1) = 2(-1)^3 - 2(-1)^2 + 3(-1) + c = 3$

$-2 - 2 - 3 + c = 3$

$-7 + c = 3 \rightarrow c = 10$

$\therefore f(x) = 2x^3 - 2x^2 + 3x + 10$ //

b. $\int f(x) = \int (6x^{\frac{1}{2}} - 10x^{-1/2})\,dx$

$= \int 6x^{1/2}\,dx - \int 10x^{-1/2}\,dx$

$= \frac{6}{3/2}x^{3/2} - \frac{10}{1/2}x^{1/2} + c$

$= 4x^{3/2} - 20x^{1/2} + c$

$f(4) = 4(4)^{3/2} - 20(4)^{1/2} + c = 2$

$4 \cdot 8 - 20 \cdot 2 + c = 2$

$32 - 40 + c = 2$

$-8 + c = 2$

$c = 10$

$f(x) = 4x^{3/2} - 20x^{1/2} + 10$ //

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c. $\int f'(x) = \int (4\sin 2x + 6\cos 3x)$

$= \int 4\sin 2x\,dx + \int 6\cos 3x\,dx$

$= 4\left(-\frac{1}{2}\cos 2x\right) + 6\left(\frac{1}{3}\sin 3x\right) + C$

$= -2\cos 2x + 2\sin 3x + C$

$f(0) = -2\cos(0) + 2\sin(0) + C = -4$

$-2 + C = -4 \rightarrow C = -2$

$f(x) = -2\cos 2x + 2\sin 3x - 2$ //

d. $\int f''(x) = \int (12x^2 - 6x + 2)\,dx$

$= \int 12x^2\,dx - \int 6x\,dx + \int 2\,dx$

$= \frac{12}{3}x^3 - \frac{6}{2}x^2 + 2x + C$

$= 4x^3 - 3x^2 + 2x + C_1$

$f'(0) = 4(0)^3 - 3(0)^2 + 2(0) + C_1 = 1$

$C_1 = 1$

$f'(x) = 4x^3 - 3x^2 + 2x + 1$

$\int f'(x) = \int (4x^3 - 3x^2 + 2x + 1)\,dx$

$= \int 4x^3\,dx - \int 3x^2\,dx + \int 2x\,dx + \int 1\,dx$

$= \frac{4}{4}x^4 - \frac{3}{3}x^3 + \frac{2}{2}x^2 + x + C_2$

$= x^4 - x^3 + x^2 + x + C_2$

$f(2) = 2^4 - 2^3 + 2^2 + 2 + C_2 = 11$

$16 - 8 + 4 + 2 + C_2 = 11$

$6 + C_2 = 11$

$C_2 = 5$

$f(x) = x^4 - x^3 + x^2 + x + 5$ //

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2. $\frac{dy}{dx} = -4x + 3$ melalui $(-1, 3)$ tentukan $f(x)$

$f'(x) = -4x + 3$

$\int f'(x) = \int (-4x + 3)\, dx$

$= -\frac{4}{2} x^2 + 3x + C$

$= -2x^2 + 3x + C$

$f(-1) = -2(-1)^2 + 3(-1) + C = 3$

$-2 - 3 + C = 3$

$-5 + C = 3$

$C = 8$

$f(x) = -2x^2 + 3x + 8$ //

3. $a(t) = (2t + 2)\ m/s^2$; $V(1) = 4\ m/s$; $S(1) = 6\ m$

$V(t) = \int a(t) = \int (2t + 2)\, dt$

$= \frac{2}{2} t^2 + 2t + C_1$

$= t^2 + 2t + C$

$V(1) = 1^2 + 2(1) + C = 4$

$1 + 2 + C = 4$

$C_1 = 1$

$V(t) = t^2 + 2t + 1$

$S(t) = \int v(t) = \int (t^2 + 2t + 1)\, dt$

$= \frac{1}{3} t^3 + t^2 + t + C$

$S(1) = \frac{1}{3}(1)^3 + 1^2 + 1 + C = 6$

$\frac{1}{3} + 1 + 1 + C = 6$

$\frac{7}{3} + C = 6 \rightarrow C = \frac{11}{3}$

$S(t) = \frac{1}{3} t^3 + t^2 + t + \frac{11}{3}$ //

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4. $v(t) = \sin 2t$, $x(t)$ jika $x(0) = 0$

$x(t) = \int v(t) = \int \sin 2t \, dt$

$= -\frac{1}{2} \cos 2t + c$

$x(0) = -\frac{1}{2} \cos(0) + c = 0$

$\therefore -1/2 + c = 0 \rightarrow c = 1/2$

$x(t) = -\frac{1}{2} \cos 2t + \frac{1}{2}$ //

5. $i(t) = \frac{E}{R}\left(1 - e^{-\frac{R}{L}t}\right)$ $q(t)$ jika $q(0) = 0$

$q(t) = \int i(t) = \int \frac{E}{R}\left(1 - e^{-\frac{R}{L}t}\right) dt$

$= \frac{E}{R} \int \left(1 - e^{-\frac{R}{L}t}\right) dt$

$= \frac{E}{R}\left(t - \frac{1}{-R/L} e^{-\frac{R}{L}t}\right) + c$

$= \frac{E}{R}\left(t + \frac{L}{R e^{\frac{R}{L}t}}\right) + c$

$q(0) = \frac{E}{R}\left(0 + \frac{L}{R}\right) + c = 0$

$\frac{EL}{R^2} + c = 0 \rightarrow c = -\frac{EL}{R^2}$

$q(t) = \frac{E}{R}\left(t + \frac{L}{R e^{\frac{R}{L}t}}\right) - \frac{EL}{R^2}$ //

$= \frac{E}{R}\left(t + \frac{L}{R e^{\frac{R}{L}t}} - \frac{L}{R}\right)$ //